

EVOLUTIONARY DISCOVERY OF REINFORCEMENT LEARNING ALGORITHMS VIA LARGE LANGUAGE MODELS

Syggkounas, Loutfi & Persson — GECCO 2026

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WHAT RL IS USED FOR

Reasoning LLMs

o1, DeepSeek-R1 — RL makes them think

LLM alignment

RLHF, DPO, GRPO, Constitutional AI

Robotics & autonomy

π_0 , OpenVLA, Atlas, Waymo, AlphaStar

Science & operations

AlphaDev, AlphaTensor, chip design, data-center cooling

This paper targets control RL — Gymnasium / MuJoCo; not LLM-RL.

RL VS. ES AT A GLANCE

Reinforcement Learning

Learn a policy by trial & error, updated with gradients on a scalar reward.

Where it runs today

- AlphaGo, AlphaStar, MuZero
- ChatGPT / Claude / Gemini (RLHF)
- o1, DeepSeek-R1 reasoning
- Humanoid locomotion, manipulation

The dominant paradigm.

Evolutionary Search

Maintain a population, mutate & recombine, keep the fittest. No gradients.

Where it runs today

- Neural architecture search (NAS)
- Hyperparameter tuning (PBT, PB2)
- Reward discovery (Eureka, REvolve)
- Non-differentiable problems (Fun-Search)

The niche tool — used where RL can't go.

THE PROBLEM WITH RL

The same handful of template families keeps getting recycled. LLM alignment has DPO / GRPO / ORPO / SimPO / KTO, but those are all preference-weighted PG. For control RL (this paper's domain), DQN '15, PPO '17, SAC '18, TD3 '18 are still the workhorses.

Sample efficiency

- PPO / SAC: 10^6 – 10^7 env steps on MuJoCo
- Humans: $\sim 10^2$ steps

Reproducibility

- $3\times$ return swings from seeds & HPs
- PPO impls disagree across codebases

Fragility

- Reward hacking, credit-assignment drift
- Deadly triad: bootstrap + FA + off-policy

Human bottleneck

- New algorithm = 1–2 yrs of PhD work
- The update rule has never been automated

THE PROBLEM WITH ES

ES is powerful but boxed in by the designer — flexible and parallel, yet stuck with the search space you wrote down.

Sample cost

- 3–10× more env steps than A3C / PPO
- No gradient, no bootstrap, no TD reuse

Dimensionality

- Stalls past $\sim 10^6$ parameters
- RL scales to billions of params

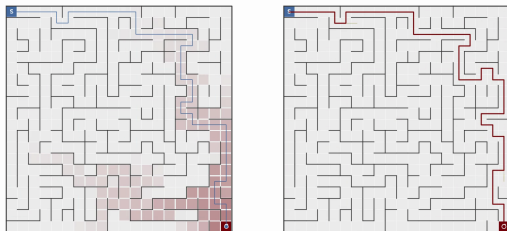
Search space

- DSL / primitives are hand-designed
- Cannot express what you didn't foresee

Blind mutations

- Random perturbations waste evals
- No semantic guidance on what to change

DEMO --- WHEN RL WINS

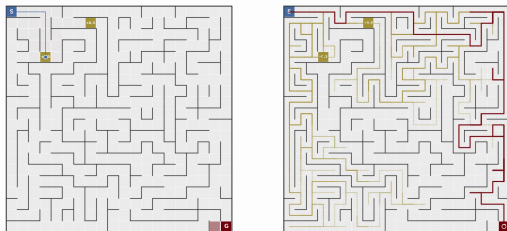


Same 20×20 maze, sparse goal reward.

- RL: 212K env steps, ~ 50 -step path
- ES: **$24\times$ more env steps** for the same result

Clean credit assignment is RL's home turf.

DEMO --- WHEN ES WINS



Add two small “trap” rewards near the start.

- RL: **0 / 2,000** episodes reach the big goal
- ES (MAP-Elites): **248 / 1,000** generations do
- 302 total breakthroughs · first at gen 44

Deceptive landscapes are where population search wins.

WHAT THIS PAPER IS

- Each candidate is an executable Python `compute_loss` function
- An LLM proposes mutations and crossovers over that code
- Each variant is scored by actually training agents in Gymnasium
- Extends REvolve — aimed at the **loss function**, not the reward
- Prompt bans actor-critic, value functions, and Bellman updates
- Discovers two new algorithms — CG-FPD and DF-CWP-CP
- Competitive with PPO and SAC across 10 Gymnasium benchmarks

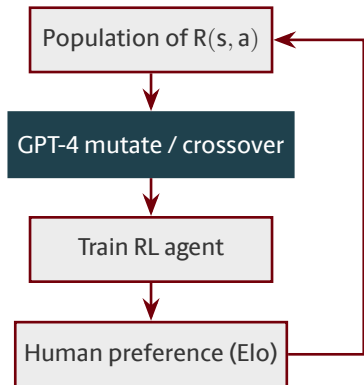
THE CANONICAL RL ALGORITHMS

- DQN (2015) — value-based, discrete action spaces
- TRPO (2015) — trust-region policy gradient
- DDPG (2015) — off-policy continuous control
- PPO (2017) — clipped policy gradient, the workhorse
- SAC (2018) — entropy-regularized off-policy
- TD3 (2018) — twin-critic DDPG

WHAT AUTOML HAS AUTOMATED

- Architecture — automatically picking network shape, depth, width
NAS, AmoebaNet, AutoML-Zero
- Hyperparameters — learning rate, batch size, clipping, γ
PB2, OptFormer, PBT
- Reward function — writing the scalar reward the agent optimizes
Eureka, REvolve
- Preference loss — the loss used to learn from pairwise preference data
Christiano RLHF, DPO, DiscoPOP, PEBBLE
- **The update rule itself** — the full RL training loop as code
open until this paper

PARENT WORK: REVOLVE (2024)



What REvolve evolves

- Executable Python **reward functions**
- Island populations, LLM variation
- Tested on driving, humanoid, dexterous hands

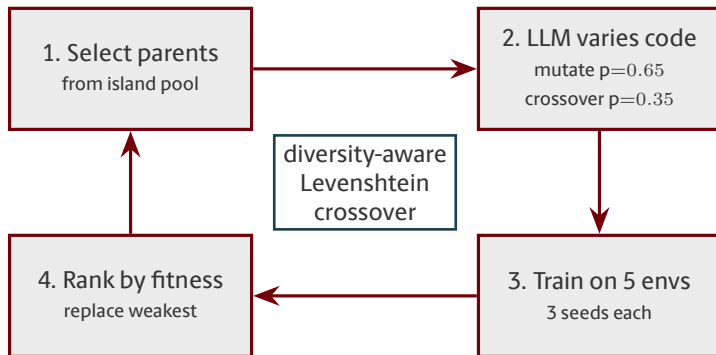
The move in this paper.
Evolve the **loss function** instead of the reward.

THE GAP THIS PAPER FILLS

What gets evolved		Representative method
Architecture	✓	NAS, AmoebaNet
Hyperparameters	✓	PB2, OptFormer
Reward function	✓	REvolve, Eureka
Preference loss (align)	✓	DiscoPOP
Update rule (DAG)	✓	Co-Reyes 2021
Update rule (black-box NN)	✓	LPG, DiscoRL
Update rule (executable Python, LLM-evolved)	★	Sygekounas et al. 2026

Everything above is context. The bottom row is **the thesis**.

THE EVOLUTIONARY LOOP



Island model · 9 individuals per island · 24 new candidates per generation.

THE TWO ALGORITHMS IT INVENTED

CG-FPD

Confidence-Guided Forward Policy Distillation

Distill short-horizon plans from a learned latent dynamics model. Planning acts as the supervised teacher signal.

No value function. No policy gradient. No Bellman.

DF-CWP-CP

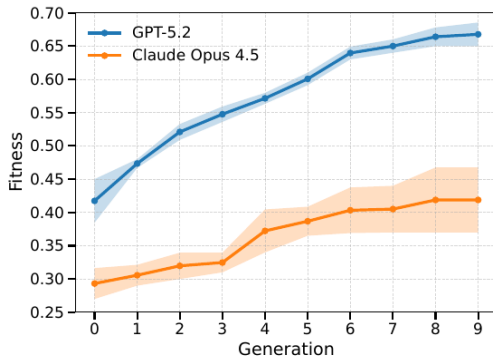
Differentiable Forward Confidence-Weighted Planning with Controllability Prior

Differentiable world-model rollouts + confidence-weighted desirability + controllability prior.

Also avoids every canonical RL mechanism.

Evolution rediscovered **model-based planning** — from scratch.

THE PROPOSER LLM MATTERS. A LOT.



- GPT-5.2 climbs to **0.67**
- Claude 4.5 Opus caps at **0.42**
- Both monotone; GPT-5.2 is sharper on every generation

Same framework, same envs, same seeds — just swap the LLM.

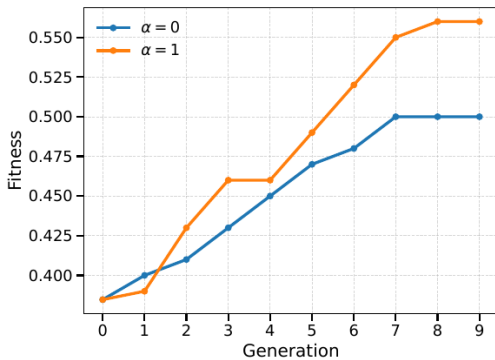
RESULTS MATCH PPO AND SAC

Environment	PPO	SAC	CG-FPD	DF-CWP-CP
CartPole	500.0	—	500.0	500.0
LunarLander	246.6	—	241.2	260.6
MountainCar	−128.1	—	−105.8	−108.6
Acrobot	−63.5	—	−90.6	−78.6
HalfCheetah	1579.1	4988.5	2407.9	2103.8
Swimmer	95.0	87.8	247.5	219.8
Walker2d	3163.2	4595.3	1603.8	1297.6
InvertedPendulum	1000.0	1000.0	1000.0	1000.0

Competitive with hand-designed baselines **across 10 environments**.

Evolved on 5 training envs; remaining 5 are generalization tests.

ABLATION: DIVERSITY WEIGHT



Levenshtein weight α

- $\alpha = 0$: no structural regularization $\Rightarrow 0.50$
- $\alpha = 1$: full regularization $\Rightarrow 0.56$
- $\alpha = 0.5$: **0.65**

Some code-similarity nudge helps;
too much or too little hurts.

STRENGTHS VS. WEAKNESSES

Strengths

- First **full update-rule** discovery via LLM
- Genuinely novel mechanisms (no value, no PG)
- Clean extension of REvolve's machinery
- LLM-HPO handles fitness fragility cleanly

Weaknesses

- 30 h per candidate on $4\times A100$
- Narrow benchmarks (CartPole / MuJoCo only)
- Is GPT-5.2 discovering or remembering?
- Only 2 evolutionary seeds per LLM
- No head-to-head vs. DiscoRL
- Constitutional constraints never ablated

THANK YOU!

Questions & Discussion

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